PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Smart Traffic Management System for Urban Congestion

A PROJECT REPORT

Submitted by

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "**Smart Traffic Management System for Urban Congestion**" is a bonafide work of "Syed Thousif(20221CSE0234), Gajjula Ashok Vardhan Reddy (20221CSE0217), Kambam Nithin Kumar Reddy (20221CSE0213)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.

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We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named Syed Thousif, G Ashok Vardhan Reddy, Kambam Nithin Kumar Reddy, hereby declare that the project work titled "**Smart Traffic Management System for Urban Congestion"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The problem solution of the project "Traffic Forecasting for Intelligent Transportation System using Machine Learning" is to revolutionize the traffic management in urban areas through predictive analytics real-time. It uses machine learning models that get trained from information gleaned out of traffic sensors, CCTV cameras, and past traffic patterns and apply these trends to predict traffic density, vehicles count, and road clearance duration for progressively precise next intervals of time. The predictive feature of the system reduces congestion, reduces travel delay, and improves decision-making in dynamic traffic control. Adding image-based parking slot detection also provides easier parking management through the provision of real-time visual data, improved accuracy, and reduced effort in manual surveillance.

The system is implemented via two main modules: the Admin Module and the User Module. The Admin Module enables system administrators to manage parking slot availability and user booking requests securely, ensuring efficient allocation and minimizing misuse. The User Module provides an easy-to-use interface to customers for signing up, getting traffic updates, and pre-reserving parking space, with real-time rescheduling facilities in times of emergencies or roadblocks. Through the convergence of machine learning, computer vision, and real-time data processing, this smart system maximizes urban transportation efficiency, improves emergency response, and facilitates smart city initiatives for planning and safer trips.

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ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
API	Application Programming Interface
GPS	Global Positioning System
UI	User Interface
UX	User Experience
DB	Database
SQL	Structured Query Language
AWS	Amazon Web Services
YOLO	You Only Look Once
CCTV	Closed Circuit Television
LSTM	Long Short-Term Memory
RF	Random Forest
DFD	Data Flow Diagram
UML	Unified Modelling Language
ER	Entity Relationship
SDG	Sustainable Development Goal
CO_2	Carbon Dioxide
JSON	JavaScript Object Notation

Chapter 1

INTRODUCTION

1.1 Problem Statement

Traffic congestion and poor parking management have become chronic and increasing problems in high-density urban environments. The growth in the number of vehicles has exceeded the growth of supporting infrastructure, and the result is congested roads, parking shortages, and widespread commuter discontent. Legacy systems are non-real time and non-predictive, and thus it becomes difficult to react to sudden traffic spikes or emergency vehicle routing. Additionally, the failure to present accurate and current information regarding parking slot vacancy leads to time wastage, higher fuel consumption, and additional pollution. Such deficiencies illustrate the need for a well-informed, automated system capable of processing the complexity of contemporary traffic patterns while providing adaptive and scalable solutions.

1.2 Objective of the Project

The core goal of this project is to design and implement a machine learning-powered smart traffic forecast and parking lot management system. It will predict traffic and parking space status from real-time and historical information. Thus, it facilitates enhanced planning and faster decision-making by frequent users and emergency responders. The system will also aid in routing emergency vehicles by predicting road clearance time and providing alternative routes. Another goal is to improve the end-user experience with a smooth interface providing live information and convenient navigation facilities. Overall, the system is aimed at minimizing urban congestion, encouraging traffic efficiency, and facilitating safer and more sustainable mobility behaviour.

1.3 Scope

This project will solve urban traffic problems using an online smart transport system. It entails developing a double-module system with an Admin Module and a User Module. The Admin Module provides city traffic officials or system administrators with the ability to manage parking slot databases, monitor traffic updates, and respond to user booking requests. The User Module does allow users to register, login, book parking lots, and acquire real-time traffic forecast and updates pertaining to emergency issues. Technical scope involves the application of machine learning algorithms to forecast traffic flow patterns, camera feed-based image-based detection of parking, and real-time traffic monitoring using traffic movement and count. The system also takes into account emergency vehicle priority and real-time routing optimization. The platform is developed using current web technologies such as React.js, Express.js, TypeScript, and MongoDB. Future scalability includes the possibility of integrating IoT devices, mobile apps, and AI-based traffic light control for city-wide deployment.

1.4 Motivation

Modern metropolises of the world are experiencing critical traffic congestion as a result of sheer density of motorized vehicles on the roads with a lack of proper parking spots. In most city centres, drivers waste hours navigating to available parking spaces, resulting in pointless jams, wastage of fuel, and emission of harmful environmental pollutants. Ambulance or fire services are even obstructed in their movement because of traffic blocks and absence of live road-clearing information.

The motivation for this project is the need to address these issues of everyday life with data-driven, cutting-edge solutions. Based on the applications of machine learning algorithms and real-time data processing, the system in this presentation has the capability to transform existing traffic and parking systems into smarter, more responsive systems. The end result is to minimize frustration, maximize safety, and create a more efficient and intelligent urban transportation experience for city planners and commuters alike.

1.5 Project Introduction

Traffic management and parking are among the most challenging problems in impairing the quality of life of modern fast-developing cities. With an increasing number of automobiles and constrained infrastructure, cities experience extreme pressure to keep their transport systems running smoothly and safely. Conventional traffic control systems not only lag behind in terms of technology but also lack the capability to respond flexibly to real-time conditions, emergencies, or varying parking space demand.

This project, "Traffic Prediction for Intelligent Transportation System using Machine Learning," is targeted at providing an industrial-strength and scalable solution to these issues. It uses real-time traffic data, image-based parking slot detection, and sophisticated machine learning algorithms to predict and manage traffic flows intelligently. The infrastructure is built on two corner modules: administrators' slot and traffic data handling module, and a users' module to locate and book parking space in return for real-time traffic information. Double-edged interaction brings in an equilateral system that is balanced in prioritizing operational efficacy as well as end-user experience. From predictive analytics and route intelligence, the project sets the foundation for smarter, safer, and cleaner transport systems for cities.

1.6 SDGs

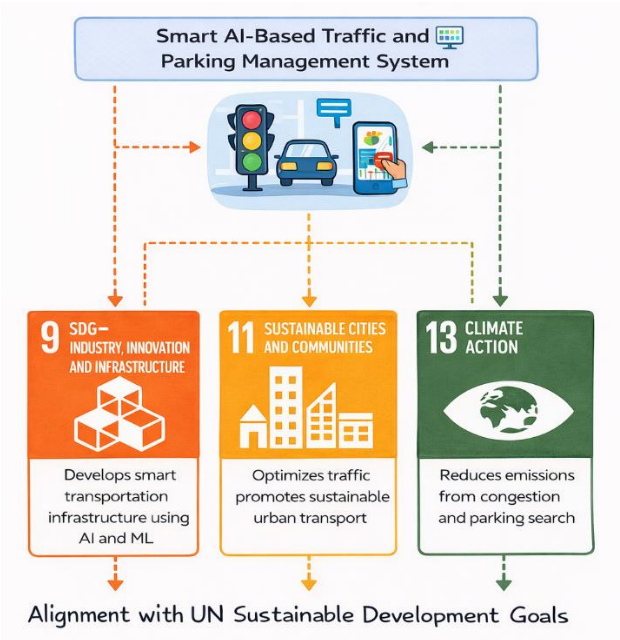

Figure 1.1 Sustainable development goals

- Align the project objectives with United Nations Sustainable Development Goals (UN SDGs) as illustrated in Fig. 1.1.
- The proposed Smart AI-Based Traffic and Parking Management System contributes to several SDGs by improving urban mobility, reducing congestion, and promoting sustainable transportation.
- Proper citation and referencing are included for the SDG framework.

Alignment with SDGs

The proposed system supports multiple Sustainable Development Goals by improving transportation efficiency and reducing environmental impact. Intelligent traffic prediction and smart parking management help cities manage traffic flow effectively and minimize unnecessary vehicle movement.

SDG 9 – Industry, Innovation and Infrastructure: The project applies artificial intelligence and machine learning technologies to develop smart transportation infrastructure, improving traffic management and supporting innovative urban mobility solutions.

SDG 11 – Sustainable Cities and Communities: By reducing traffic congestion, optimizing signal timings, and providing real-time parking information, the system contributes to safer, more efficient, and sustainable urban transportation systems.

SDG 13 – Climate Action: The system reduces vehicle idling and unnecessary travel caused by congestion or parking search, which leads to lower fuel consumption and reduced CO_2 emissions, supporting climate action initiatives.

1.7 Overview OF Project Report

This project report presents the design and implementation of a "**Smart AI-Based Traffic and Parking Management System"** developed to improve urban traffic flow and parking efficiency.

Chapter 1 introduces the project by presenting the problem statement, objectives, scope, motivation, and an overall introduction to the system.

Chapter 2 reviews related research and existing studies in the domain of intelligent transportation systems and smart traffic management.

Chapter 3 discusses the research gaps identified in existing approaches, including limitations in real-time precision, insufficient integration of diverse data sources, poor handling of unstructured data, lack of user-centric features, inadequate emergency vehicle prioritization, scalability issues, and limited adaptability to sudden events.

Chapter 4 explains the proposed methodology by describing the existing system, its disadvantages, and the proposed smart traffic and parking management framework along with its benefits and implementation approach.

Chapter 5 details the objectives of the system, including real-time traffic prediction using machine learning, smart parking slot detection through computer vision, emergency vehicle routing, and an integrated platform for efficient traffic and parking management.

Chapter 6 focuses on system design and implementation, presenting input and output design along with various UML diagrams such as class, sequence, activity, deployment, component, ER, and data flow diagrams.

Chapter 7 outlines the project execution timeline using a Gantt chart covering requirement analysis, UI/UX design, backend and frontend development, testing, deployment, and maintenance.

Chapter 8 presents the expected outcomes and benefits of the system, including improved traffic flow efficiency, smart parking management, enhanced user convenience, and accurate real-time traffic prediction.

Chapter 9 discusses the system results and implementation outputs such as parking slot booking, slot information management, and traffic forecasting modules.

Chapter 10 concludes the report by summarizing the project contributions and overall system performance.

Finally, **Chapter 11** lists the references used in the study, and **Chapter 12** provides additional supporting materials in the appendix.

Chapter 2

LITERATURE REVIEW

In traffic congestion control system based on an emergency-aware reinforcement learning and a non-dominated sorting genetic algorithm were suggested by Al-Heety et al. (2026) [1]. Their algorithm learns adaptive Q-learning and optimized communication infrastructure to dynamically tune the traffic lights and give priority to emergency vehicles. The system has shown that it is efficient in traffic flow, decreased delays before traffic, and emergencies in urban traffic with multi-interception.

Another proposed next-generation smart traffic management system that is anticipated to alleviate congestion in Indian metropolitan cities was proposed by Goswami and Kadao (2025) [2]. Their platform combines IoT sensors, artificial intelligence, and machine learning algorithms to display traffic status and actively regulate the timings of the signals. Predictive analytics and adaptive signal control in the model considerably minimized travel time and congestion levels and emissions of traffic.

Xu and Li (2025) suggested a real-time forecast-traffic congestion model, which relies on dynamic risk field modeling and the multi-source data fusion [3]. They use a combination of deep learning methods and models of physically explainable risk fields to understand spatiotemporal traffic patterns. The framework makes correct predictions of congestion and helps in the proactive management of traffic systems of complex urban networks.

The paper by Prathiba et al. (2025) proposed a real-time traffic and power grid control system with a digital twin enabling it to work in smart cities [4]. A strategy based on predictive analytics, real-time data processing, and the dual-level digital twin architecture is employed to achieve simulations of urban infrastructure behavior. By optimally distributing resources and decision making, the system made the traffic flow more efficient and decreased the congestion.

Wairagade et al. (2025) came up with an AI-based smart traffic management system that can help alleviate enhancing urban congestion through machine learning and computer vision technology [5]. Their system gathers real-time traffic data between sensors and cameras with the help of the IoT and than processes it with the help of AI algorithms to make signal durations change dynamically. The strategy contributed to the efficiency of traffic flow, consumption of less fuel, and emergency vehicle movement.

A proposal was made (Naramdeo et al., 2025) in the form of an intelligent traffic management system that dynamically regulates the traffic lights according to the number of vehicles and the sensor that detects emergency vehicles on the road [6]. Their system involves sensor-based vehicle detection, and communication networks to ensure traffic density is monitored, and adjust the timings of signals on-the-fly. Simulation outcome has demonstrated the effectiveness on traffic flow performance as compared to traditional setting of fixed time signals.

Laanaoui et al. (2024) proposed the real-time anomaly detection and load balancing system of urban traffic control [7]. They involve their algorithm based on machine learning and parallel processing of data to predict the traffic density and identify abnormal traffic situations. The system enhances traffic management by offering optimum route guidance and alleviating the traffic jams in urban roadways.

Raghunath et al. (2024) proposed a traffic supply and traffic control mechanism based on Temporal Convolutional Networks with Federated Learning [8]. They have a decentralized approach which enables several traffic sensors and cameras to analyze traffic data collectively without exchanging raw information. The system enhances prediction accuracy of congestion and ensures privacy of data in distributed traffic monitoring systems.

Fadila et al. (2024) engaged in an in-depth literature review of technologies of smart urban traffic management within the framework of the Fourth Industrial Revolution [9]. Their work emphasized the need to combine the use of artificial intelligence, IoT, blockchain, and intelligent transportation system to enhance urban mobility and sustainable transportation management.

Wang et al. (2024) suggested a cyber-physical system structure to control the intersections of the roadways in the city based on a Model- Based Systems Engineering [10]. Their architecture combines the elements of sensing, communication, and computation so as to facilitate the efficient monitoring and control of traffic lights. The system enhances coordination and reliability of traffic on complex intersection settings.

Xu et al. (2023) have designed an urban traffic system cloud-based digital twin platform, which provides an opportunity to have real-time situational awareness and cyber-physical control of the system [11]. They have a system that combines both IoT sensor and sophisticated analytics to give the smart city transportation infrastructure traffic monitoring, predictive analytics, and signal optimization.

Jin et al. (2022) online suggested an agent-based traffic recommendation system as an efficient method of strategic traffic management of the city [12]. Their system involves a multi-agent architecture to dynamically produce operational strategies to control traffic at cross-road. The method allows making traffic decisions automatically, and can coordinate intersection better.

The system developed by Oliveira et al. (2021) to fully monitor a traffic control system in real-time is based on wireless communication networks [13]. Their system can create centralized monitoring and dynamic control over traffic lights in various intersections which enhances better coordination by the traffic signals and reduces the congestion.

In an article by Nagapragna and Joshi, the authors suggested a smart-traffic management system whereby intersections are detected by using Arduino and infrared sensors [14]. They have an embedded system that dynamically controls traffic signals in response to vehicle density and offer a cost effective and scalable solution to enhancing traffic flow in cities.

Lanke and Koul (2013) presented a smart traffic management system (RFID-based) that was to dynamically manage traffic lights depending on the frequency of vehicles being detected [15]. Their system employs RFID tagging and controllers that recognize vehicles and control timing of the signals based on them, allowing identification of congestions early and better management of the traffic.

Summary of Literatures reviewed

Table 2.1 Summary of Literature reviews

Author(s) & Year	**Title / Study Focus**	**Methodology / Framework**	**Limitations / Gaps**	**Relevance to Our Project**
Al-Heety et al. (2026)	Emergency-aware traffic congestion control system	Reinforcement learning with non-dominated sorting genetic algorithm for adaptive signal control	Requires advanced communication infrastructure	Supports emergency vehicle prioritization and adaptive traffic control
Goswami and Kadao (2025)	Smart traffic management system for Indian metropolitan cities	IoT sensors with AI and ML for predictive analytics and adaptive signal timing	Limited integration with parking systems	Provides foundation for AI-based congestion prediction

Xu and Li (2025)	Real-time traffic congestion prediction model	Multi-source data fusion with deep learning and dynamic risk field modeling	High computational complexity	Helps improve congestion prediction accuracy
Prathiba et al. (2025)	Digital twin–based traffic and power grid control system	Predictive analytics with real-time data processing and digital twin architecture	Implementation complexity in real environments	Demonstrates smart city infrastructure integration
Wairagade et al. (2025)	AI-based smart traffic management system	Machine learning and computer vision using IoT sensors and cameras	Requires large-scale data collection	Relevant for traffic monitoring and dynamic signal optimization
Naramdeo et al. (2025)	Intelligent traffic light control system	Sensor-based vehicle detection with dynamic signal timing	Limited scalability in large cities	Supports real-time traffic density monitoring
Laanaoui et al. (2024)	Real-time anomaly detection in urban traffic	Machine learning with parallel data processing	Limited integration with emergency routing	Helps detect abnormal traffic situations
Raghunath et al. (2024)	Traffic prediction using TCN and federated learning	Temporal convolutional networks with distributed data processing	Complex implementation and high computational load	Improves privacy-preserving traffic monitoring
Fadila et al. (2024)	Smart urban traffic management technologies review	Integration of AI, IoT, blockchain, and intelligent transportation systems	Mostly theoretical discussion	Provides technological background for smart traffic systems
Wang et al. (2024)	Cyber-physical traffic intersection control system	Model-Based Systems Engineering integrating sensing, communication, and computation	Implementation challenges in real-world deployment	Supports intelligent intersection management

Xu et al. (2023)	Cloud-based digital twin traffic system	IoT sensors with cloud analytics for real-time monitoring and signal optimization	Requires high data processing capability	Relevant for smart city traffic monitoring
Jin et al. (2022)	Agent-based traffic recommendation system	Multi-agent architecture for adaptive traffic control strategies	Limited real-world deployment	Demonstrates automated traffic decision-making
Oliveira et al. (2021)	Wireless communication-based traffic monitoring system	Centralized monitoring of traffic signals using wireless networks	Infrastructure dependency	Supports coordinated traffic signal management
Nagapragna and Joshi (2017)	Arduino-based smart traffic system	Embedded system using IR sensors to detect vehicle density	Limited scalability and functionality	Shows low-cost traffic signal automation
Lanke and Koul (2013)	RFID-based smart traffic management	RFID vehicle detection for dynamic signal timing	Requires RFID infrastructure on vehicles	Early approach to intelligent traffic management

Table 2.1 summarizes the literature reviewed for the development of the Smart AI-Based Traffic and Parking Management System. The reviewed studies highlight different approaches used in intelligent transportation systems, including artificial intelligence, machine learning, IoT, and computer vision techniques. Many researchers have focused on traffic congestion prediction, adaptive signal control, and real-time traffic monitoring using sensor and camera data. Some works also explored reinforcement learning, digital twin technology, and multi-agent systems to improve traffic management. However, several limitations were identified, such as lack of parking integration, high computational complexity, and insufficient handling of real-time multi-source data. In addition, many systems do not effectively prioritize emergency vehicles or provide user-centric features. Based on these research gaps, the proposed system aims to integrate traffic prediction, smart parking detection, and emergency vehicle routing into a unified intelligent traffic management platform.

Chapter 3

RESEARCH GAPS OF EXISTING METHODS

3.1 Limitations in Real-Time Precision

The current systems of traffic prediction fail to give any real-time accurate prediction as they are based on obsolete or incomplete sources of data. Most of the models depend on static historical data, which is no longer valid in an era of dynamic changing traffic conditions as those caused by accidents, weather conditions, or arbitrary incidents. Such absence of real-time updateability cripples the efficacy of the dynamic urban life.

3.2 Insufficient Integration of Diverse Data Sources

While planning traffic prediction, non-integration of few disparate sources of data such as that from live camera videos, GPS location, social media updates, and readings from IoT sensors does not give the overall picture of traffic behavior. Heterogeneous data integration is very much required for traffic forecasting to be context-aware and accurate, which is most often missing in many methods.

3.3 Poor Handling of Unstructured Data

There is a huge difference in how these existing models process unstructured data such as video or images. Most of the applications make use of structured data like speed measurements, traffic volume, etc. Very few systems, however, use the visual data for applications like parking detection or incident detection. Little work has been carried out primarily in advanced image processing and applications of deep learning to unstructured data.

3.4 Absence of User-Centric or Personalization Features

Most traffic systems focus at macro-level and do not offer user-specific prediction or routing for individual based on. There is, therefore, a growing need for developing models that are very much user specific in terms of demand such as preferred paths, availability of parking spaces near destinations, and estimation of travel time based on personal trends.

3.5 Neglect of Emergency Vehicle Concern

Most of the existing methods do not give priority for the emergency vehicles while drawing traffic management schemes. The consequences of delayed emergency response would be catastrophic in situations where there is high road congestion. There is, however, a significant but undeveloped area in this area of research directing how predictive models factor and give priority to emergency vehicles.

3.6 Scalability and Deployment Difficulties

Most of the machine learning models designed in the laboratory are not at all scalable for real cities. High computation costs, not cloud-compatible, and therefore infeasible for large-scale city data hinder making them available for large-scale deployment. The research needs to focus on lightweight-scalable models that can be deployed in a large city.

3.7 Limited Adaptability to Sudden Events

Traffic models in existence are usually helpless when it comes to sudden changes, such as protests, road closures, or flooding, which totally transform traffic patterns and escape the attention of most predictive models. The development of such resilient models that will adapt quickly and learn about such interruptions is still necessary.

Chapter 4

METHODOLOGY

4.1 Existing System

The traffic and parking management infrastructures of today are highly dependent on static rule-based systems that are not responsive enough to modify themselves based on real-time scenarios. These systems typically comprise manual procedures, such as allocating parking lots or managing road jamming, which are not only time-consuming but also prone to human error. Furthermore, the conventional systems are not coupled with real-time traffic data, rendering it challenging to react in real time to varying traffic conditions, particularly in urban areas where traffic is highly complex and dynamic.

In addition, current systems are inferior in handling emergency cases. For instance, they have no intelligence to expedite the entry of ambulances or fire trucks, resulting in emergency response latency. Consequently, such systems are responsible for ineffective use of resources, traffic congestion.

4.2 Disadvantages of the Current System

1. Static Management

There is no real-time adaptability of the existing system. Allocation of traffic and parking spots is according to pre-programmed rules, and the system is inflexible and unable to address live traffic patterns or parking demand changes.

2. No Predictive Capabilities

Legacy systems are reactive, not proactive. They do not predict congestion, traffic volumes, or parking availability through predictive analysis, hence inefficient traffic flow and poor planning.

3. Limited Emergency Response

There is no system to give priority to emergency vehicles like ambulances or fire trucks. This is due to a limitation that causes delayed emergency responses and puts lives at risk.

4. Higher Congestion Levels

Lack of live traffic integration makes the system ineffective in rerouting vehicles during peak periods or surprise roadblocks, leading to excessive traffic jams.

5. Increased Driver Frustration

Commuters suffer due to lack of good traffic information and inefficient parking management. This leads to longer commute durations and irritation, especially in urban areas with dense traffic.

4.3 Proposed System

The system designed is an intelligent traffic and parking management system that is machine learning and real-time information-based for providing proactive solutions to urban mobility issues. This system is optimized to offer live traffic prediction, image-based car park detection, dynamic slot allocation, and routing priority-based traffic for ambulances, etc.

The system is architected into two major modules:

I. Admin Module

This module is specifically for the system administrators operating the backend activities.

Log in safely and verify the dashboard. Track and manage parking bay statistics. Approve or cancel bookings. Deal with emergency vehicle movement and guide them best. Handle analytical reports on system use, congestion level, and performance ratios.

II. User Module

The user module offers end-users such as commuters or drivers services.

Register and log in into the system. See actual time parking availability. Book parking slots on the basis of actual-time prediction. See traffic information, emergency messages, and navigation recommendation.

See individual recommendation based on past usage and prevailing traffic pattern. This segment ensures administrators and users both have an intuitive and effective experience, optimized for their individual needs and role.

4.4 Proposed System Benefits

- **Real-Time Insights**
  The system gathers and processes real-time traffic to inform users of road conditions at the moment and correct parking spot availability, allowing for more informed decisions.
- **Improved Efficiency**
  By dynamically adapting to traffic, flow, and slot availability, the system minimizes idle time, waiting times, and congestion.
- **Improved Emergency Handling**
  Emergency responders are afforded priority via specialized algorithms that determine the quickest and least congested routes in real-time, leading to faster response times.
- **User-Friendly Experience**
  Through a responsive and clutter-free user interface, the platform facilitates easy searching, reservation, and finding of parking slots with little inconvenience.
- **Data-Driven Approach**
  The system exploits machine learning to keep learning from traffic data, user activities, and road conditions, thus enhancing the accuracy of prediction with time.

4.5 Proposed Methodology

1. Data Collection and Preprocessing
   The system starts with the collection of various types of traffic-related data:

- Live traffic feed from GPS, roadside sensors, and CCTV cameras.
- Traffic data for the past for pattern analysis.
- Parking lot data from image feed for assessing availability.

All this information is pre-processed to normalize and clean the input. Unimportant or redundant information is eliminated, and missing values are processed to supply good quality data for training machine learning models.

2 Machine Learning Model Development
The system's wisdom lies in the predictive capability. For this:

  - Supervised learning algorithms such as Random Forest, LSTM (Long Short-Term Memory), and Decision Trees are trained to provide predictions of traffic density, congestion level, and expected parking availability.

- Time-series processing is employed to understand traffic flow over time.
- Feature engineering is employed to gain more indicators helpful for improving the performance of the model.

3. Image-Based Slot Detection

   Computer vision techniques are used to find the occupancy of parking spaces. Some of the key techniques used are:

   - YOLO (You Only Look Once) for object detection in real-time.
   - OpenCV for edge detection and image processing.
   - The modules work live camera streams to find out if a parking slot is occupied or not, and provide the user interface with real-time feedback.

4. Module Integration

   Admin and user modules are brought into a core system that:

   - Speaks to the ML-based prediction engine.
   - Talks to the database for booking and status updates.
   - Provides user action synchronization and admin control.
   - It enables smooth data transfer and ensures that system behaviour is uniform across all interfaces.

5. Emergency Vehicle Routing

   Upon the detection of an emergency vehicle, the system automatically performs the following:

   - Scans traffic along various routes.
   - Determines the shortest and most vacant route.
   - Dynamic rerouting of regular vehicles to make way for the emergency vehicle.
   - Not only does it improve response time but also road safety for all passengers.

6. Real-Time Updates and Alerts

   Users are constantly alerted via the mobile or web application for:

   - Unforeseen traffic congestion.
   - Updated estimated time of arrival adjustments.
   - Number of parking spaces available.
   - Advisories related to emergencies.

- The feature maintains users informed and responsive to changing conditions.

7. Feedback Loop and Model Refinement

The system incorporates a ceaseless learning loop where:

- User action, booking record, and real-time feedback are gathered.
- ML models are retrained at intervals to capture new patterns.
- The overall prediction engine adapts to accommodate hitherto unknown traffic scenarios more accurately.

Chapter 5

ANALYSIS AND DESIGN

5.1 Objectives of Smart Traffic and Parking Management System

The urban landscape of today is changing at breakneck rates, but traffic congestion and bad parking continue to be the preeminent issues of cities worldwide. Existing nonadaptive systems have limited capability to respond adaptively to real-time variations in traffic streams, parking availability, and emergency routing. To meet these urgent needs, the Smart Traffic and Parking Management System outlined here leverages machine learning capability, real- time processing, and smart routing to deliver a state-of-the-art, adaptive system. This report establishes the system's primary objectives to mitigate congestion, enable parking availability, and prioritize emergency response, making urban mobility a smarter, safer experience.

5.1.1 Objective 1: Integrated Platform for Traffic and Parking Management Overview

The system will be a centralized system for intelligent parking allocation and real-time traffic management. The system integrates live data streams, smart decision-making, and convenient services to improve urban mobility.

Problems Solved

- Traffic and parking systems independently operating make coordination inferior.
- Underutilization and inefficiency result from manual or static allocation of parking space.
- Failure to integrate results in sluggish or second-best choices at congestion hour.
- Smart System Solution
- Traffic and parking integrated in real-time on a single platform.
- Predictive analysis to direct drivers to the closest available parking.
- Admin dashboard to track slot usage and approve bookings.

Benefits

- Smooth user experience and instant decision-making.
- Optimal use of resources through dynamic allocation.
- Decreased search time for parking spaces and decreased traffic load.

5.1.2 Objective 2: Real-Time Traffic Forecasting Using Machine Learning

Introduction

Predictive and real-time analytics are crucial in managing traffic congestion in urban areas. The system uses ML algorithms to forecast traffic and optimize flow.

Problems Addressed

Ineffective route planning resulting from inability to forecast traffic buildups. Static systems lack the ability to cope with dynamic traffic patterns.

Solution

- Machine learning models (Random Forest, LSTM) learning from real-time and historic data.
- Real-time congestion levels with recommended alternative routes.
- Peak hours and possible bottlenecks prediction.

Benefits

- Supports planning and routing to the future.
- Reduces fuel consumption and travel time.
- Supports urban planners to look at traffic flow to plan infrastructure.

5.1.3 Objective 3: Smart Parking Slot Detection Using Computer Vision

Overview

Free parking space real-time detection is responsible for reducing driver frustration and city traffic. The system employs computer vision, deep learning, and image processing to provide automatic slot detection.

Problem Statement

- Drivers waste time searching for parking.
- Monitoring parking areas manually is labor-intensive and error-prone. Smart System Solution
- Computer vision algorithms (e.g., YOLO, OpenCV) scan real-time CCTV feeds to detect free and occupied slots.
- Real-time availability on user dashboards. Admin page for slot status validation and update.

Advantages

- Shortened parking search time.
- Improved utilization of parking spaces.
- Low-burdensome auto-monitoring system.

5.1.4 Objective 4: Routing of Emergency Vehicles with High Priority

Background

Time is of the essence in dealing with emergencies. Legacy systems lack dynamic diversion or emergency responder road clearing.

Problems Solved

Emergency vehicle congestion due to a lack of proper dynamic prioritization. Delays in clearing and absence of real-time notification of drivers.

Solutions

- Dynamic emergency paths computed based on predicted ML and real-time traffic data.
- Warning system to alert immediate users to yield and not drive on certain paths.
- Admin override feature for immediate de-congesting of the route.
- Benefits
- Improved emergency response time and security.
- Less disruption of emergency operations.
- Improved coordination of system users' and service providers.

5.1.5 Objective 5: Smart Interface for Improved User Experience

Overview

High adoption and usage-friendly design is made possible. The system offers web and mobile interfaces for searching and real-time booking.

Challenges Overcome

- Massive systems drive people away from their utilization.
- Personalized messages within the system at the wrong time result in decreased efficiency.

Solution

- Streamlined UI for login, user registration, booking of parking slot, and traffic notification.
- Traffic notice, slot confirmation notice of the slots, and emergency notice.
- Admin dashboard for system management and booking control.

Benefits

- Simplified user experience for ease.
- Enables user trust and engagement.
- Faster decisions and transactions.

5.1.6 Objective 6: Continuous Improvement through Feedback Loops Background

Systems need to continually improve so that they are effective and relevant. Feedback loops help to maximize performance.

Problems Solved

- Static systems worsen over time because they do not learn.
- Low responsiveness and customization.
- Smart System Solution
- Traffic, activity, and slot activity are tracked and measured.
- Machine learning algorithms are updated and retrained on new data regularly.
- Users and admins can provide explicit feedback so that predictions can be refined.

Pros

- Enhanced traffic and parking prediction.
- History-based personalized user recommendation.
- The mentioned objectives complement one another to create a wise, scalable, and effective parking and traffic management system. Being equipped with real-time monitoring, predictive
- analytical insight, self-configuring routing, and user-centered interfaces, the platform not only remedies the contemporary urban disease but adapts itself with it.

Explanation

Traditional parking and traffic management systems are static and operate using fixed rules, which makes them less effective as traffic conditions change over time. These systems lack adaptability, responsiveness, and personalization for users. To overcome these limitations, a smart system is proposed that continuously monitors traffic flow, parking availability, and user activity using real-time data. Machine learning algorithms analyze this data and are regularly updated to improve prediction accuracy. User and administrator feedback is also incorporated to refine system performance. As a result, the system provides improved traffic and parking predictions along with personalized recommendations for users. By integrating real-time monitoring, predictive analytics, adaptive routing, and user-friendly interfaces, the platform offers a scalable and efficient solution for modern urban traffic and parking management.

Chapter 6

SYSTEM DESIGN & IMPLEMENTATION

6.1 Key Points to Consider Input Devices:

Input and Output Design Input Design Introduction Input design is a very important step in the system development cycle. It establishes the way information is input to the system in such a manner as to ensure accuracy, efficiency, and usability. As the quality of input has a direct bearing on the quality of output, input form, screen, and control designs need to adhere to strict guidelines that minimize user mistakes and maximize data processing.

The software can take input from any device like PCs, Magnetic Ink Character Recognition (MICR), Optical Mark Recognition (OMR), touch interface, etc. Design Principles: Input forms and screens should be: Specific and effective for data retrieval and recording. Precise and reliable. Easy to understand and complete. Consistent in appearance with emphasis on simplicity and readability. Eased in terms of the way users are likely to interact with UI components.

Goals of Input Design:

- To design effective data output processes.
- To reduce the amount of output data.
- To generate output documents or record data automatically.
- To build formatted output data records, interfaces, and forms.
- To use validation rules and output controls to ensure data integrity.

6.2 Output Design

Output design deals with the way output is delivered following processing. It aims to provide correct, on-time, and useful information to users. Format, content, and media of the output need to match the requirements of the end-user.

Objectives of Output Design: In order to make outputs work properly and remove redundant data. In order to meet different expectations and needs of different user roles. In order to control the quantity and speed of outputs. In order to display outputs in a suitable format and deliver them to right recipients (e.g., print, mail, on-screen presentation). To provide outputs in real-time, supporting real- time decision-making.

6.2.1 UML Diagrams

Unified Modelling Language (UML) is applied for visualizing, specifying, building, and documenting the elements of a software system. The following are the major UML diagrams applicable to the project: Use Case Diagram Use case diagrams represent system features as seen by end users (actors). They emphasize: System boundaries Actors (Admin, User) Use cases (Login, Book Slot, View Traffic, etc.) Relationships (associations, generalizations, includes, extends) Use case diagrams are crucial for user interaction and system response understanding.

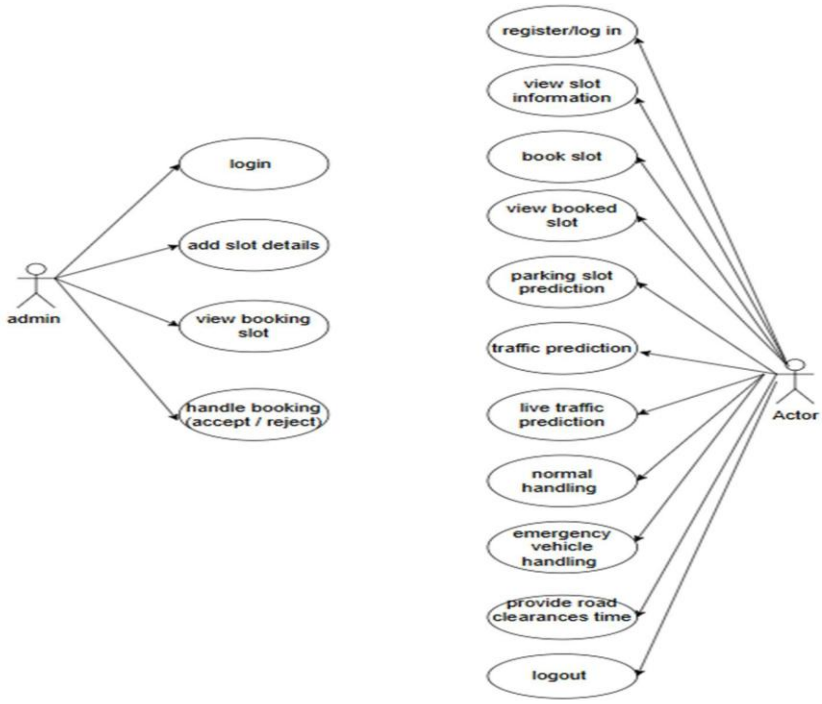

Figure 6.1: UML Diagram

6.2.2 Class Diagram

Class diagrams give a static picture of the system, describing: Classes (e.g., Admin, User, Booking, Slot, Traffic) Attributes (e.g., userID, slotID, bookingTime) Methods (e.g., addSlot (), bookSlot (), getTrafficData ()) Relationships like associations, inheritance, and aggregation This diagram is useful for object identification and their relations at implementation time.

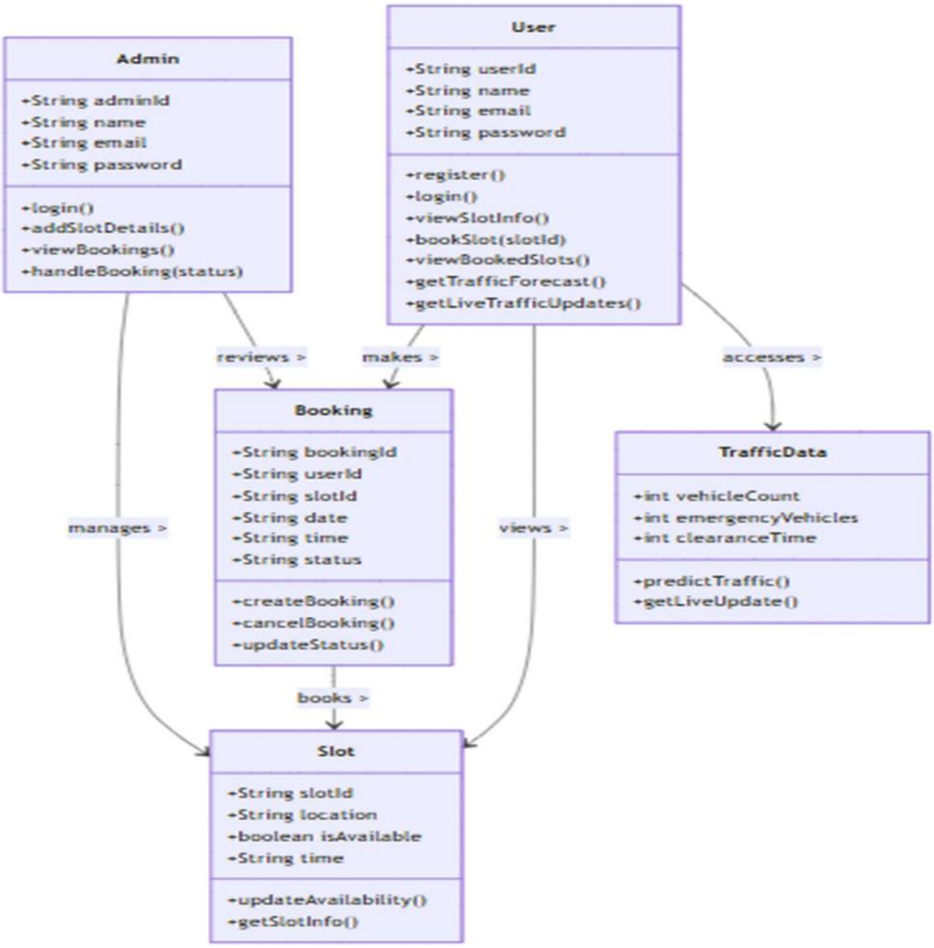

Figure 6.2: Class Diagram

6.2.3 Sequence Diagram:

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a

Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

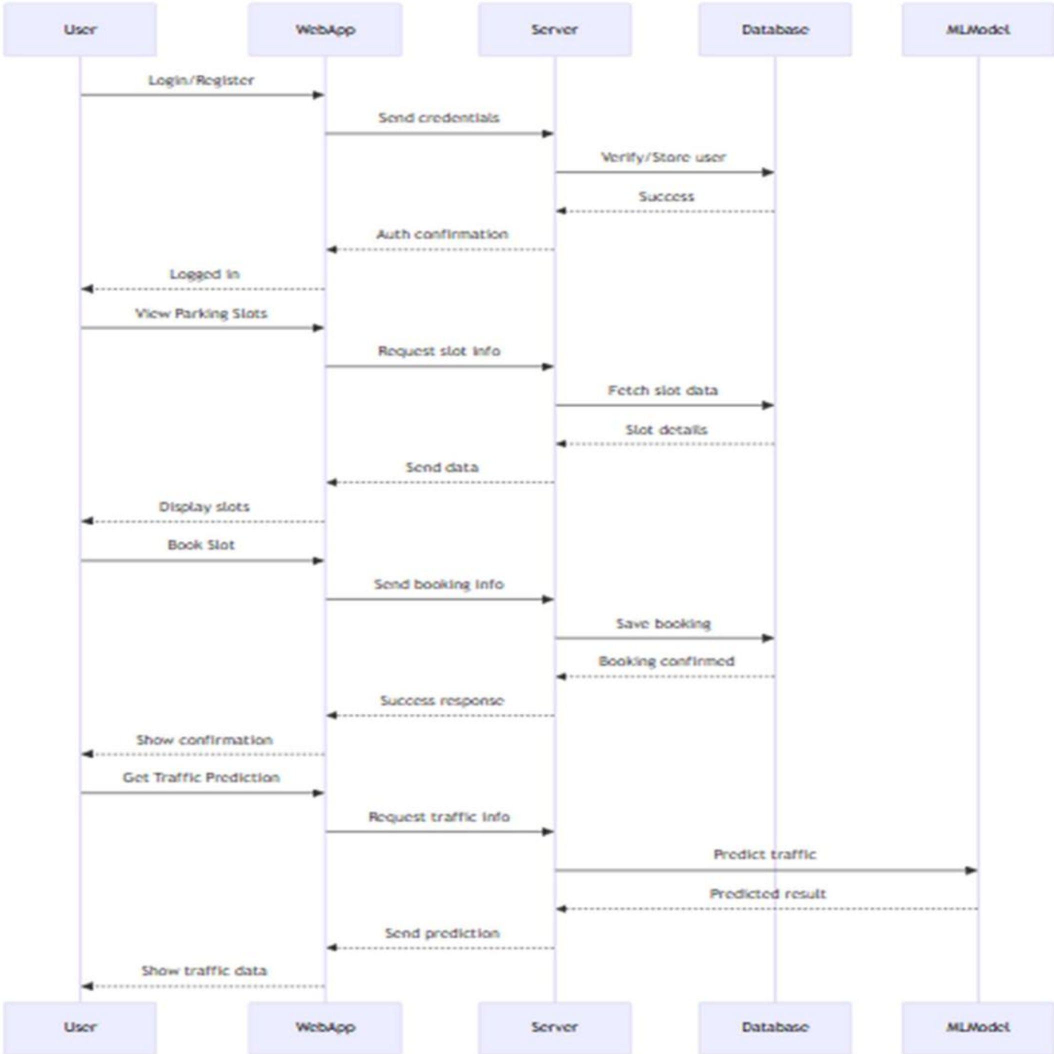

Figure 6.3: Sequence Diagram

6.2.4 Collaboration Diagram

Collaboration diagrams stress object connections and message exchange. In contrast to sequence diagrams, they concentrate on object placement and interaction routes. Numbered arrows mark method invocation, illustrating: How objects collaborate to perform tasks Communication sequence and structure

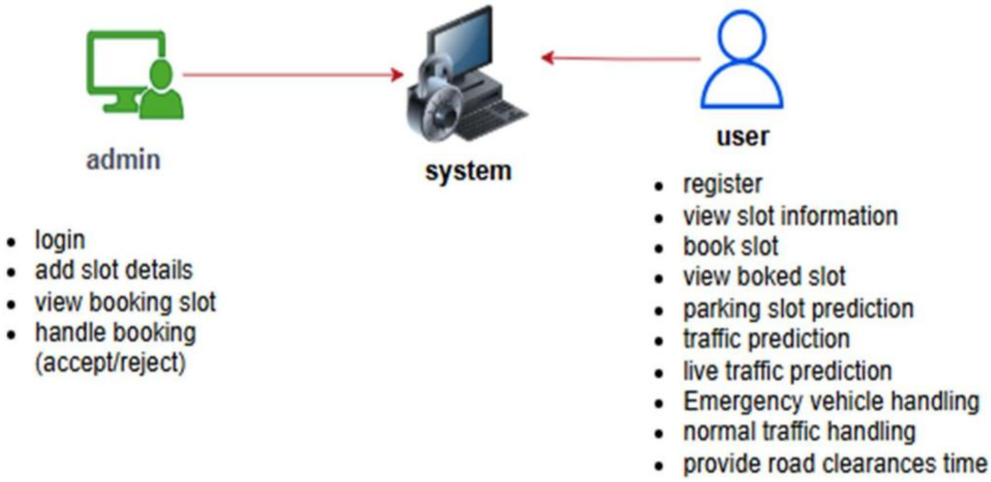

Figure 6.4: Collaboration Diagram

6.2.5 Deployment Diagram

Deployment diagrams illustrate the physical deployment of artifacts (software elements) onto nodes of hardware. This comprises: Servers (Application Server, Database Server) Devices (User Mobile, Admin PC) Communication links and deployed software components Helpful in capturing system scalability and deployment planning

Figure 6.5: Deployment Diagram

6.2.6 Activity Diagram

Activity diagrams show the control flow between activities and decisions. They contain: Start and end nodes Decision points, parallel nodes Actions standing for system functions They assist in workflow description and process automation.

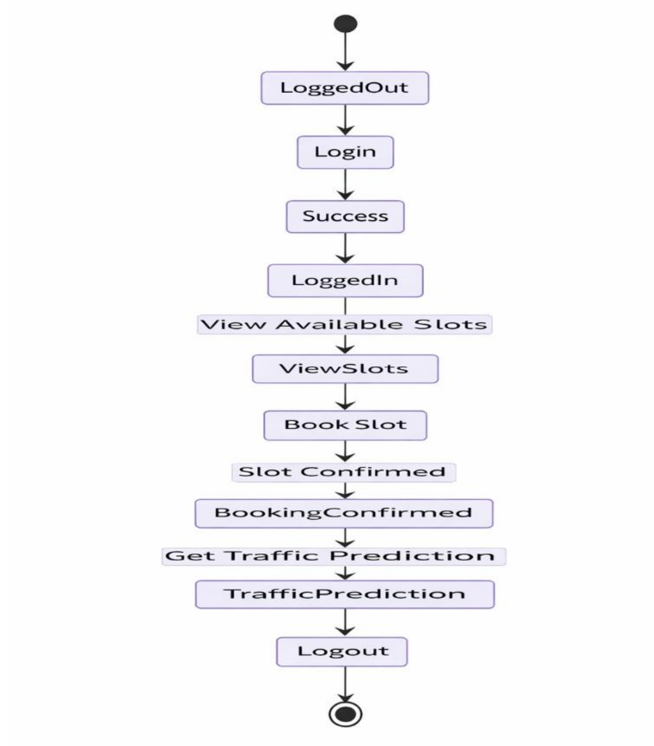

Figure 6.6: Activity Diagram

6.2.7 Component Diagram

Component diagrams represent the structure of code modules: Components (User Interface, Admin Panel, Traffic API) Interfaces given and used Inter-module dependencies This ensures all functionality needed is modularized and properly connected.

Figure 6.7: Component Diagram

6.2.8 ER Diagram (Entity-Relationship Diagram)

ER diagrams represent the database schema graphically. The main constituents are: Entities: User, Admin, Booking, Slot, Traffic Data Attributes: name, email, bookingTime, location

Relationships: Users book Slots, Admin manages Bookings, etc. These are employed to develop relational databases.

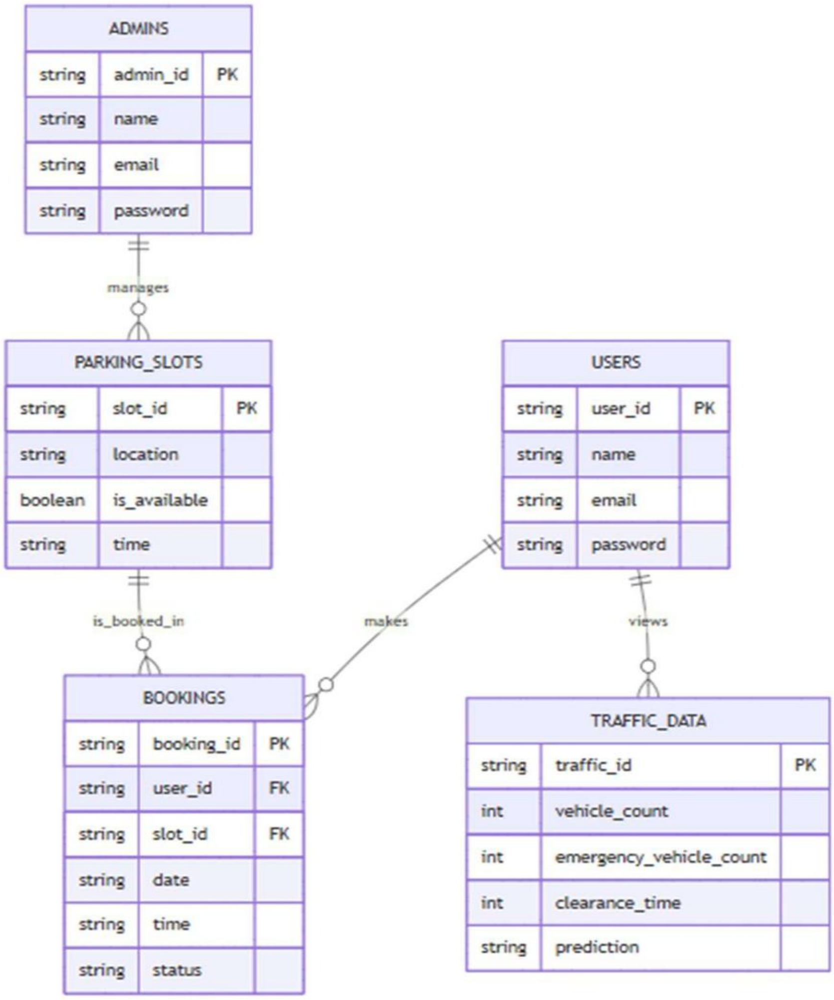

Figure 6.8: ER Diagram

6.2.9 Data Flow Diagrams (DFD)

DFDs show data flow within the system, such as sources, processes, data stores, and data destinations.

DFD Level 1: The top-level diagram has: Processes such as User Registration, Login, Slot Booking, and Traffic Prediction. Data stores such as User DB, Slot DB, and Booking DB. External entities such as Admin, User, and Traffic API.

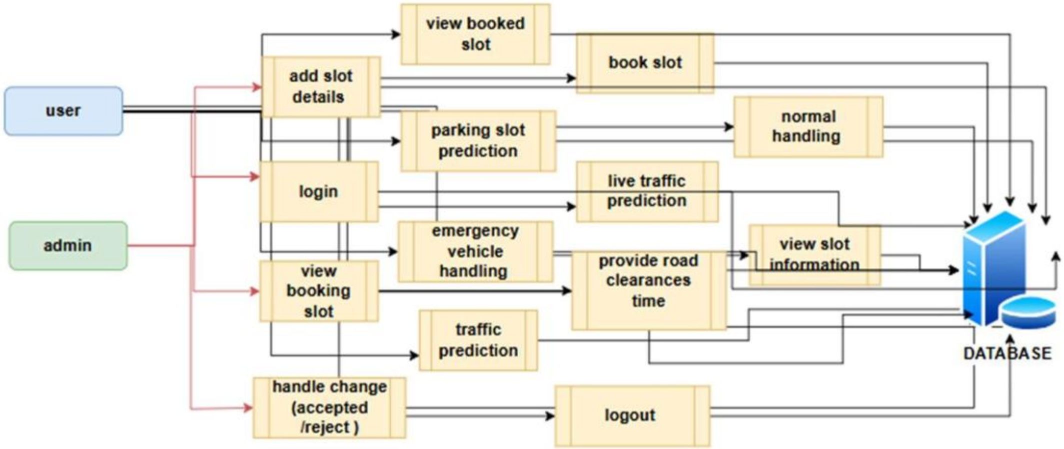

Figure 6.9: DFD Diagram

DFD Level 2: At this level, there is a more detailed decomposition: Internal processing of slot booking, validation, and availability checks. Steps to validate traffic prediction and emergency response data. Interaction with prediction-based ML models and image-based APIs. Functional Modules

1. **Admin Module**

Admin Login: Secure administration login through authentication credentials. Add Slot Details: Input location, slot ID, and current availability status.

View Booking Slots: Display all present user bookings for admin management. Handle Booking Requests: Admin can approve or reject booking depending upon availability of slot **and priority.**

2. **User Module**

User Registration and Login:

- Secure users to register and log in to the site.
- View Slot Information: Show list of available slots with location and time filter. Book Parking Slot: Form-based interactive parking booking by time and location.
- View Booked Slots: Current and historical bookings visible to users.
- Parking Slot Prediction (API-Based Image Analysis): Real-time predictions of available slots based on visual information.
- Traffic Forecasting: Forecast traffic information provided by machine learning models.
- Live Traffic Prediction: Prediction in real-time based on vehicle volume.

- Emergency Vehicle & Clearance Handling Emergency routing high-priority data processing and estimated clearance times.

Table 6.1 Functional modules

Test Case ID	Test Scenario	Precondition	Test Steps	Expected Result
TC001	User Registration	User is on the registration page	1. Open registration page2. Enter valid details3. Click **Register**	User account is created and a confirmation message is displayed
TC002	User Login	User is already registered	1. Open login page2. Enter valid email and password3. Click **Login**	User is successfully redirected to the dashboard
TC003	View Available Parking Slots	User is logged in	1. Navigate to **View Slots** page2. Request slot data	Available parking slots are displayed to the user
TC004	Book a Parking Slot	User is logged in and parking slots are available	1. Select a parking slot2. Enter booking time3. Click **Book Slot**	Parking slot is booked successfully and confirmation message is displayed
TC005	Admin Adds Parking Slot	Admin user is logged in	1. Navigate to **Add Slot** page2. Enter slot details3. Click **Add**	New parking slot is added and appears in the slot list
TC006	Admin Accepts Booking	A user has already made a booking request	1. Go to **Booking Requests**2. Click **Accept** on pending booking	Booking status is updated to **Accepted**
TC007	Traffic Prediction Request	User is logged in	1. Navigate to **Traffic Forecast** page2. Click **Get Prediction**	Traffic prediction results are displayed using real-time data
TC008	Live Traffic for Emergency Handling	Emergency vehicle data is available in the system	1. Open **Live Traffic** module2. Enable emergency mode3. View updated route and clearance information	Emergency route and traffic clearance time are displayed

Chapter 7

TIMELINE FOR EXECUTION OF PROJECT

7.1 Define Requirements:

This is the initial phase that involves gathering and analyzing the project requirements and goals. It provides the foundation for the rest of the project by establishing a clear picture of user needs, functionality, and goals.

Timeline: Phase One

7.2 Design UI/UX:

Is responsible for creating the interface and user experience such that both are aesthetically pleasing as well as simple and intuitive to utilize by farmers and stakeholders. Wireframes and prototypes can be created here.

Timeline: Post Requirement Finalization

7.3 Develop Backend:

Includes coding the project's fundamental functionality, i.e., server logic, database schema, and APIs. Technologies may include Django and MySQL, as discussed in our project stack. Timeline: After Design Phase

7.4 Develop Frontend:

Frontend is developed in order to get an interactive and working user interface. The procedure includes implementing the designed UI/UX and rolling it out together with the backend.

Timeline: Simultaneous with Backend Deployment Integrate Components:

Entails the assembly of both the frontend and the backend components to a unified framework. Confirms that from booking for labor service to leasing equipment all just works seamlessly. Timeline: Post-Development

7.5 Testing

Testing is conducted to identify and fix bugs and make the application secure, stable, and perform at its optimum level under different conditions. Functional, usability, and performance tests are carried out.

Timeline: Post-Integration

7.6 Deployment

The application runs on a live environment (e.g., cloud hosting through AWS or Firebase). The process allows access to use the project by the target group.

Timeline: Post-Successful Testing

7.7 Post-Deployment Support

Has ongoing maintenance, updates, and improvements based on users' recommendations. New features can also be implemented to further enhance the application.

Timeline: Ongoing Phase After Deployment

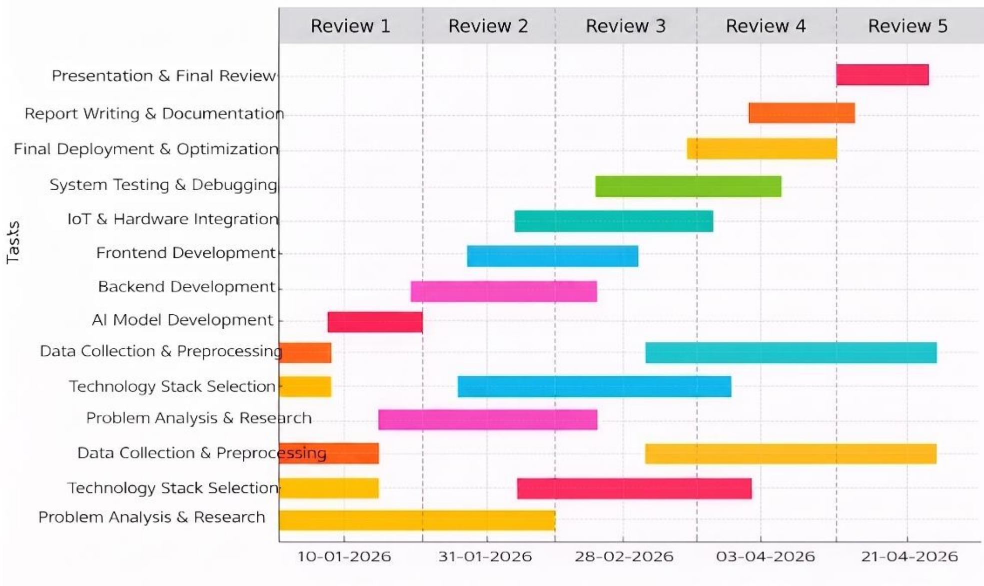

Figure 7.1: Gantt Chart

Chapter 8

OUTCOMES

8.1 Improved Traffic Flow Efficiency:

The most significant impact of the project was the substantial enhancement in traffic flow management. With the help of machine learning models to forecast traffic congestion before it occurs, users were able to make more informed travel choices like rerouting or rescheduling their journey. This predictive approach reduced delays in travel and minimized congestion in congested city centers. Consequently, city transit systems became more efficient, and vehicular flows through key intersections became less disordered and more predictable.

8.2 Smart Parking Management System:

The implementation of a smart parking system transformed the process of how users identified and booked parking spaces. By enabling both the users and administrators to interact with real-time data regarding the availability of parking slots, the system reduced the time spent searching for parking. Options like real-time slot prediction through image processing helped to ensure that users had access to the latest data, eliminating unnecessary traffic brought about by cars driving around in search of parking.

8.3 Administrator Control and Operation Transparency

For administrators, the project provided a very efficient control panel that brought everything parking and booking-related under one umbrella. Admins were able to add new slots with ease, view pending bookings, and approve or deny user requests depending on availability. Central control reduced man error, ensured smooth functioning, and enabled instant decision- making. The admin module ensured system transparency and accountability as all activity and booking history were tracked and traceable with ease.

8.4 User Convenience and Satisfaction Enhanced

The website enhanced the user experience greatly with a well-smoothed, easy-to-use registration, login, booking, and slot viewing interface. It pleased users that they could track bookings and live traffic updates using their mobile or web interface.

It helped to enable advance booking, instant traffic forecast, and availability notification that enabled overall user satisfaction to increase. This convenience also encouraged greater public adoption of smart transportation solutions.

8.5 Features with High Accuracy of Real-Time Traffic Prediction

One of the most notable results of the project was the application of machine learning algorithms for real-time traffic prediction. Past data, real-time feeds, and sensor readings from traffic sensors were fed into these models to come up with highly accurate predictions of traffic congestion levels. Over time, the model became more accurate since it was constantly learning and being retrained with new information. This aspect was one of the significant contributory elements to delay reduction, emergency response planning, and enhanced responsiveness of city infrastructure systems.

8.6 Emergency Vehicle Prioritization and Safety Assurance

One of the system's biggest success stories was prioritizing emergency service vehicles such as fire trucks and ambulances in times of crisis. The project came with an added module to track emergency vehicle movement and shift traffic forecasts based on it. In addition to making response time shorter for emergency services, this ensured public safety also improved. It cleared out other cars and forecast road clearing times, making it so emergency services would get to where they were headed more quickly with little resistance.

8.7 Basis for Future Integration as a Smart City

The project also laid a strong architecture and technology base for subsequent smart city projects to expand upon. Its extensibility due to being built on frameworks such as Django, MySQL, and cloud hosting vendors like AWS made it highly extendable to be augmented with further functionality such as auto-billing, pollution monitoring, or IoT traffic control.

8.8 Data-Driven Decision Making and Predictive Analytics

Finally, one of the long-term effects of the project is city planning based on data. With the traffic information and parking collected and analyzed in real-time, the system chooses to concentrate and notify city planners and government authorities on traffic movement, peak congestion hours of traffic, and user behavior. By doing so, the project addresses not only immediate transportation requirements but also future strategic development objectives.

Chapter 9

RESULTS AND DISCUSSIONS

9.1 Add slot details

Implementation of the Admin Module in the intelligent transportation system brought about efficient parking slot and booking management. Admins could log in securely by using authenticated credentials, so that only authorized individuals could operate the system. The "Add Slot Details" feature enabled the admin to enter key information like slot ID, location, and availability status, which were updated in real-time to the users. This centralized management interface significantly increased slot update efficiency and reduced slot data inconsistencies.

9.2 View booking slots

Using the "View Booking Slots" feature, the admin enjoyed full visibility of all the incoming booking requests. Visibility made handling the peak hours of traffic and working beforehand easy. With the "Handle Booking Requests" option, the admin was able to accept or deny user requests in real-time based on availability. Double bookings were avoided, and optimal utilization of the parking infrastructure was achieved. The outcome was a properly managed backend that formed the basis for robust user-side performance.

9.3 View slot information

The User Module too was made very interactive and safe right from the start of "User Registration and Login". Registration was easy and fast, letting the users easily register themselves with secure login protocols safeguarding their information. While logged in, the users even had the feature of "View Slot Information," which displayed the availability of parking slots and locations in a well-understandable manner. On this basis, the users would be in a position of making justified decisions without seeking a parking space.

9.4 Book parking slot

The "Book Parking Slot" function was strong and easy to operate: using this function "Book Parking Slot", one was able to book his or her slot in advance, especially the preferred time and location. The "View Booked Slots page" showed the current and previous bookings, thereby making one manage the parking history in proper order effectively. Such openness dramatically cut down on confusion and led to greater user satisfaction.

9.5 Parking slot prediction via api

The most groundbreaking of its features was the "Parking Slot Prediction via API Image" whereby image processing determined real-time availability of parking slots. The module employed computer vision techniques and sensor information to analyze real-time parking feed. Its results showed that it was highly accurate in predictions during daytime. It saved user search time and also guided them away from busy lots, increasing overall system efficiency.

9.6 Traffic forecasting

In traffic management, the "Traffic Forecasting" module, with machine learning technology, correctly predicted future traffic jams from real-time and historical data. Decision trees and linear regression models were employed by the system to predict traffic flow. Users were notified of anticipated delays, and they were able to divert themselves to other routes or change travel plans. The more information, the better the prediction, which proved that the model learns and improves on its own in the long run.

Live Traffic Forecasting was a key module that served real-time feedback based on the density of the vehicles, indication of the traffic signals, and detection of an emergency vehicle. The module fed real-time responses to the users in terms of a visually information-rich dashboard. Displaying dynamic road conditions in real-time contributed to optimal routing optimization and facilitated urban mobility. Additionally, live feedback enabled city planners to quickly react to traffic jams or accidents.

The Emergency Vehicle & Clearance Handling feature was crucial in preventing life-saving support services from being held up. The system optimized routes for ambulances and fire trucks by adapting forecasted traffic flows and determining road clearance time through intelligent algorithms. The module improved public safety by maximizing signal control and suggesting alternative routes for normal users to clear the way for emergencies. In general, the results showed that incorporating machine learning, real-time data processing, and smart user/admin interfaces made the system a complete solution for contemporary urban transport issues.

Chapter 10
CONCLUSION

The "**Smart Traffic Management System for Urban Congestion**" project delivers a novel and scalable solution for the vital issues that arise within urban transportation systems, namely those concerning traffic jams and inefficient parking systems. While cities grow in size and people use more and more vehicles, traditional approaches towards traffic management become unable to generate timely and dynamic solutions. This project utilizes the strength of machine learning algorithms to process real-time and historical traffic data, providing accurate and dependable traffic forecasting. The system's predictive ability helps city planners and commuters plan more effectively, avoid traffic-congested routes, and enhance road safety.

Some of the salient features of the system include prioritization of emergency vehicles, best routing and signal priority to be given to ambulances, fire trucks, and police vehicles while responding to emergencies. It helps to curb emergency response time and increase public safety significantly. Urban parking conditions are also dealt with in this project by leveraging a smart parking management system through detection and optimization of parking space using predictive analytics. This not only helps drivers locate parking more effectively but also discourages unnecessary car movement to search for parking, thus the use of less fuel and reduced emissions.

The system is well-planned with distinct modules for users and administrators, ensuring ease of use while providing maximum control and observation to the authorities. Users are able to obtain traffic forecasts, receive parking recommendations, and have real-time route optimization, while administrators are able to observe gross traffic pattern, assign emergency priorities, and control system parameters. The incorporation of machine learning enables the system to learn and evolve over time based on new information, improving efficiency and accuracy.

With the integration of smart automation and real-time response, this project contributes heavily towards urban mobility transformation. It establishes the overall notion of smart city development through setting a strong technology foundation for managing future traffic and infrastructure. The system not only suggests a sustainable and scalable model for the future city but also encourages decision-making, sustainability, and quality of life for cities.

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BASE PAPER

Xu, H., Berres, A., Yoginath, S.B., Sorensen, H., Nugent, P.J., Severino, J., Tennille, S.A., Moore, A., Jones, W. and Sanyal, J., 2023. Smart mobility in the cloud: Enabling real-time situational awareness and cyber-physical control through a digital twin for traffic. *IEEE Transactions on Intelligent Transportation Systems*, *24*(3), pp.3145-3156.

Available at: https://www.osti.gov/pages/servlets/purl/1958153

APPENDIX

i. Publication

4th International Conference on Data Analytics and Insights : Submission (370) has been created. Inbox ×

Summarize this email

Microsoft CMT <noreply@msr-cmt.org>
to me

Hello,

The following submission has been created.

Track Name: Wireless and Sensor Communication Systems

Paper ID: 370

Paper Title: Smart AI-Based Traffic and Parking Management System Using Machine Learning and Computer Vision

Abstract:
Traffic congestion and difficulties in parking have become common problems in contemporary cities due to the rapid urbanization. The conventional traffic control systems are dynamic and do not reflect the actual traffic conditions. The present paper suggests a Smart AI-Based Traffic and Parking Management System which will combine the concepts of machine learning and computer vision to forecast traffic jams and find vacant parking spots in real-time. The system relies on the algorithm of the Random Forest to compare historical and real time traffic data, as a result of which intelligent traffic movement forecasts and adaptive traffic lights control are possible. The use of computer vision can be used to track the parking space and aid an early reservation of parking. The suggested system is more efficient in traffic, it lowers down congestion, the travel time is also minimized and emergency vehicles are also prioritized. The solution offers scalable and cost-effective solution to smart city transportation systems.

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Authors:
- syedthousif498@gmail.com (Primary)

Primary Subject Area: Computer Vision /Intelligent Transportation Systems (ITS) Smart Cities / Smart Transportation Intern

Secondary Subject Areas: Not Entered

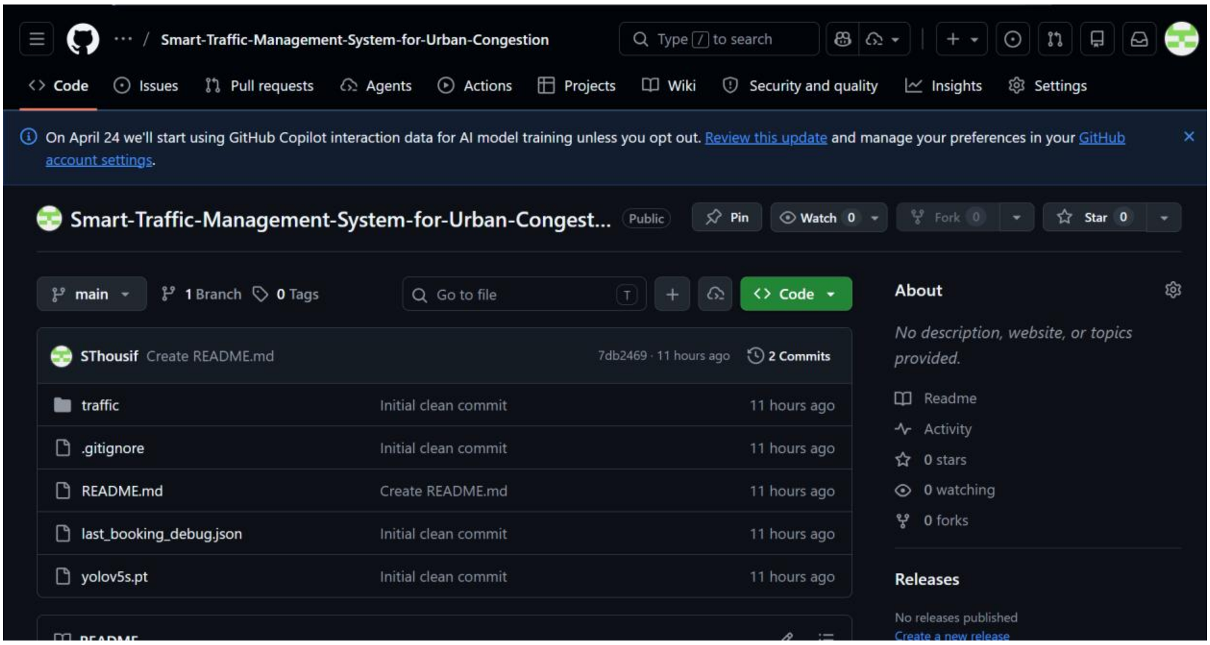
Submission Files:
Project_Paper.pdf (478 Kb, Tue, 21 Apr 2026 03:27:43 GMT)

Submission Questions Response: Not Entered

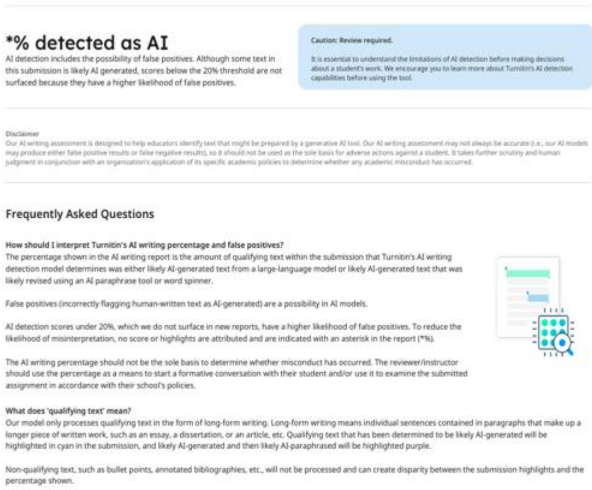
Thanks,
CMT team.

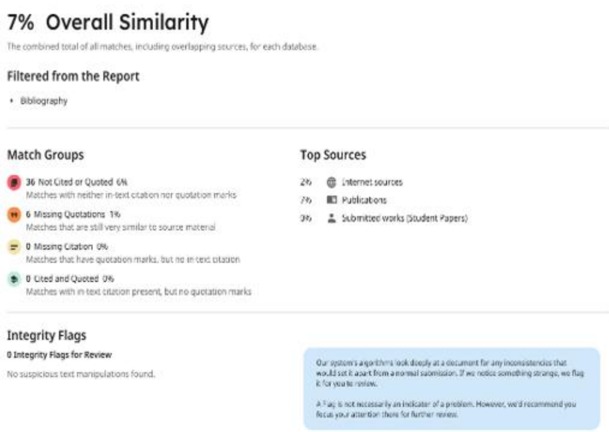
ii. GitHub Link

https://github.com/SThousif/Smart-Traffic-Management-System-for-Urban-Congestion



iii. Similarity, Plagiarism Check

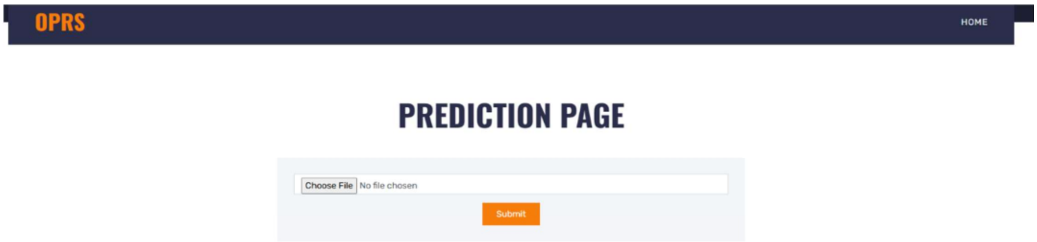




iv. Few images of Project

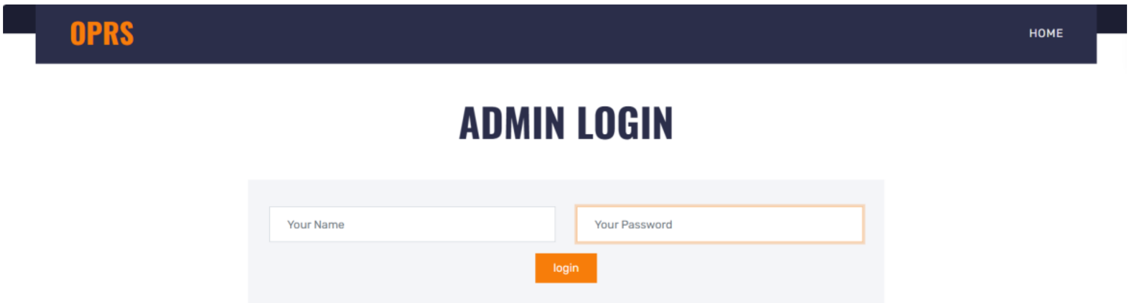
Home page: The main landing page providing navigation to parking and traffic prediction features.

Vacant parking place prediction: Interface displaying predicted available parking slots based on current data.

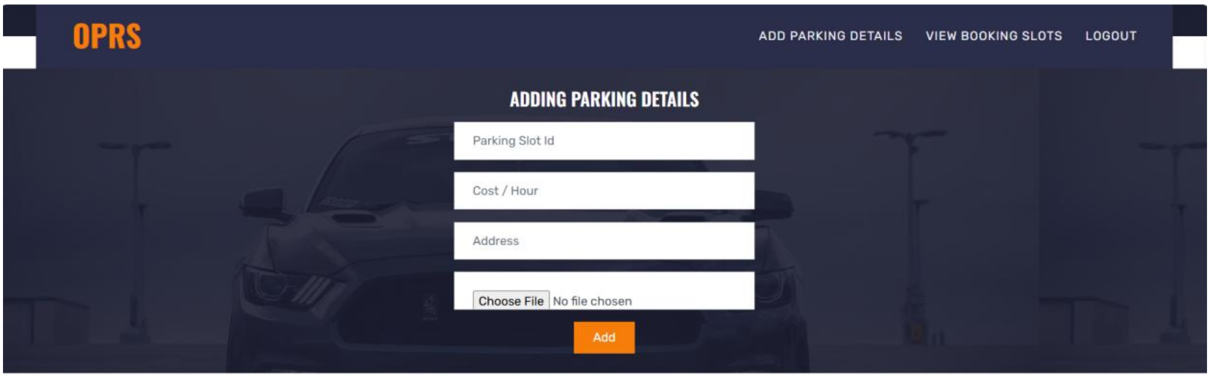


Prediction results: Page showing output of the parking vacancy prediction model with slot availability.

Admin login page: Secure login interface for administrators to access management functionalities.



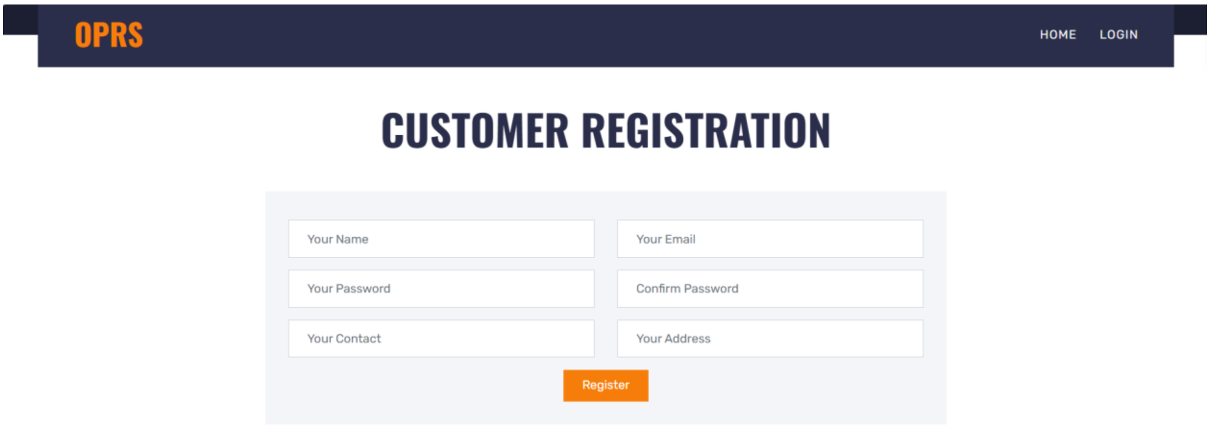
Add slot page: Form allowing admins to add new parking slot details into the system.



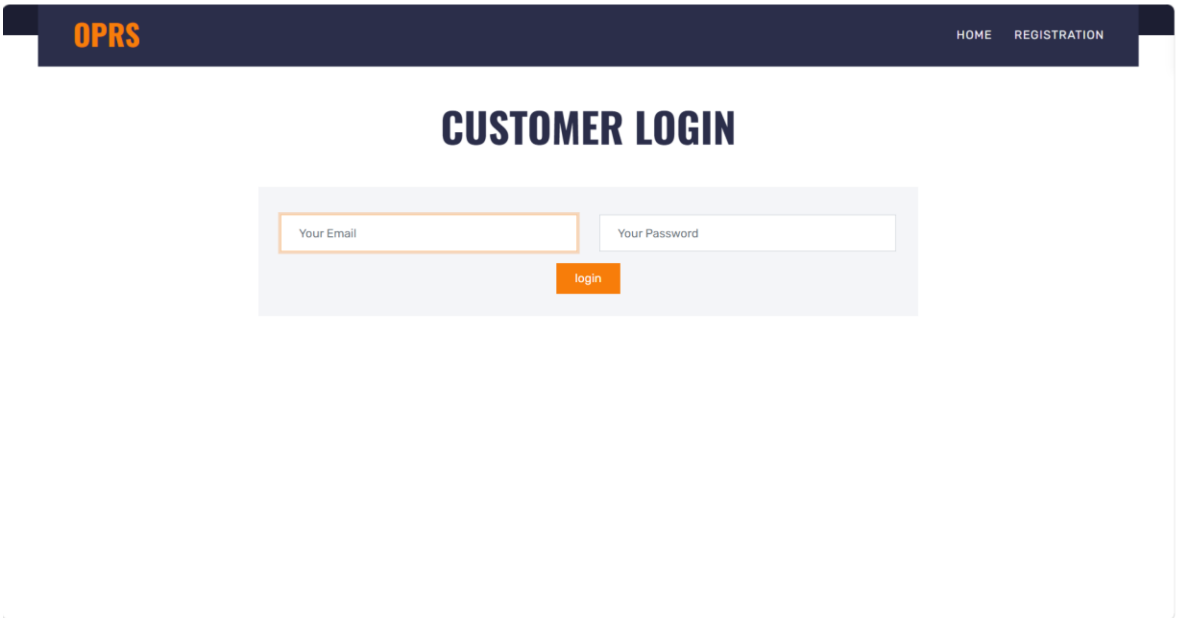
Booking slot details: Displays user's reserved parking slots with SlotID, hour cost, name on card, total hours, total amount, Action and location info.

id	slotid	hourcost	nameoncard	totalhours	totalamount	status	useremail	Action
12	D4	40	canara	23	920	locked	syedthousif498@gmail.com	accept/ reject / unlock slot
13	B4	30	Syed Riyaz	5	150	accepted	syedthousif498@gmail.com	accept/ reject / unlock slot

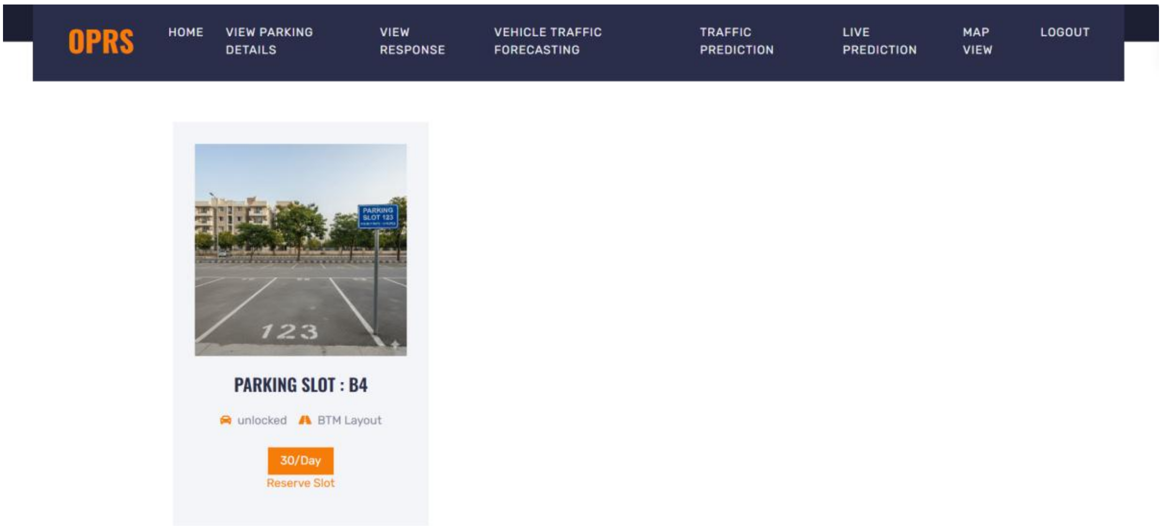
Customer registration: Sign-up form for new users to create an account in the system.



Login page: User login interface to access parking and traffic prediction services.

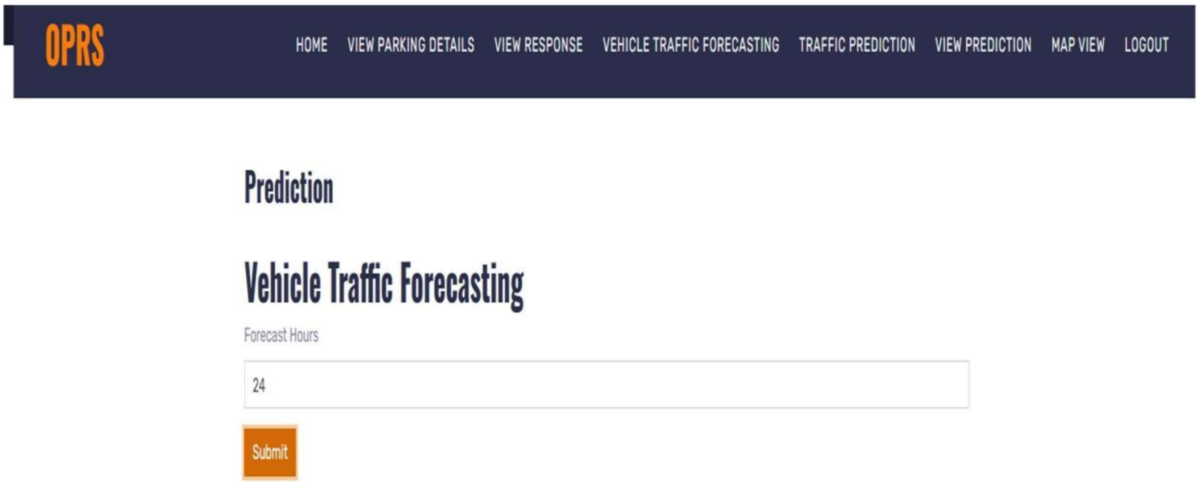


View parking slot details: Shows detailed information for each parking slot, including availability.

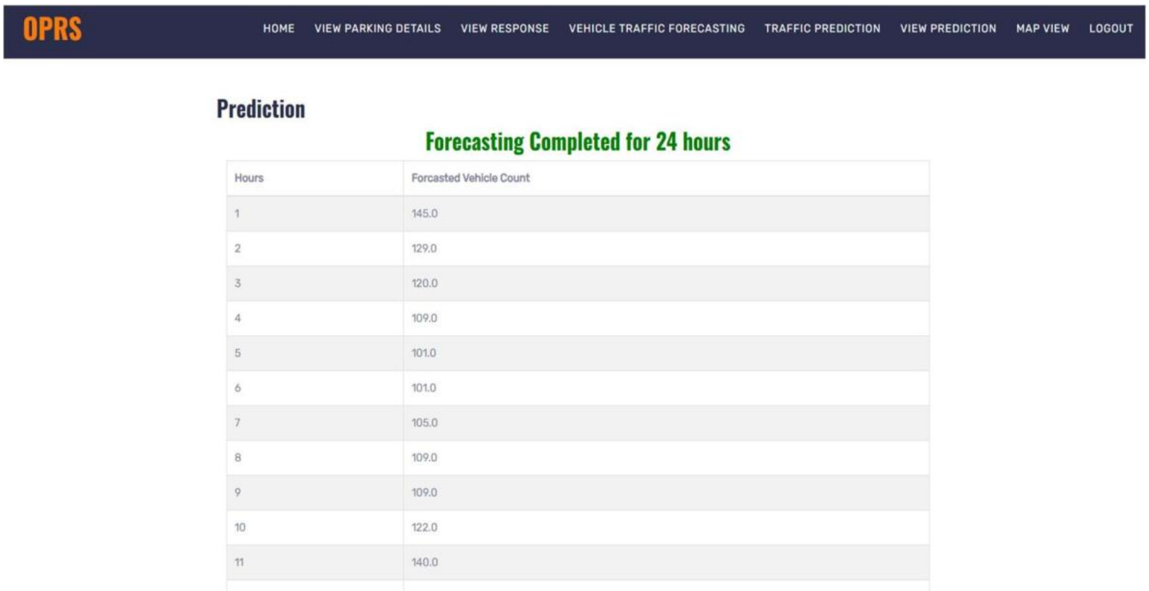


View slot response: Displays confirmation and details of the user's slot selection or reservation.

Traffic forecasting page: Interface to input parameters and initiate traffic flow prediction.

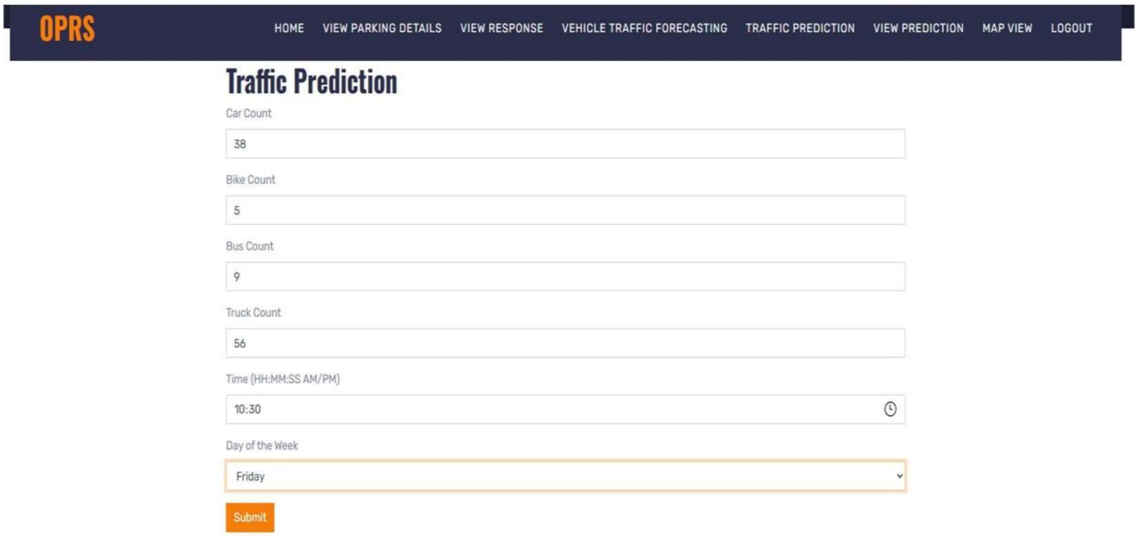


Forecasting details: Results page showing predicted traffic trends over selected timeframes.

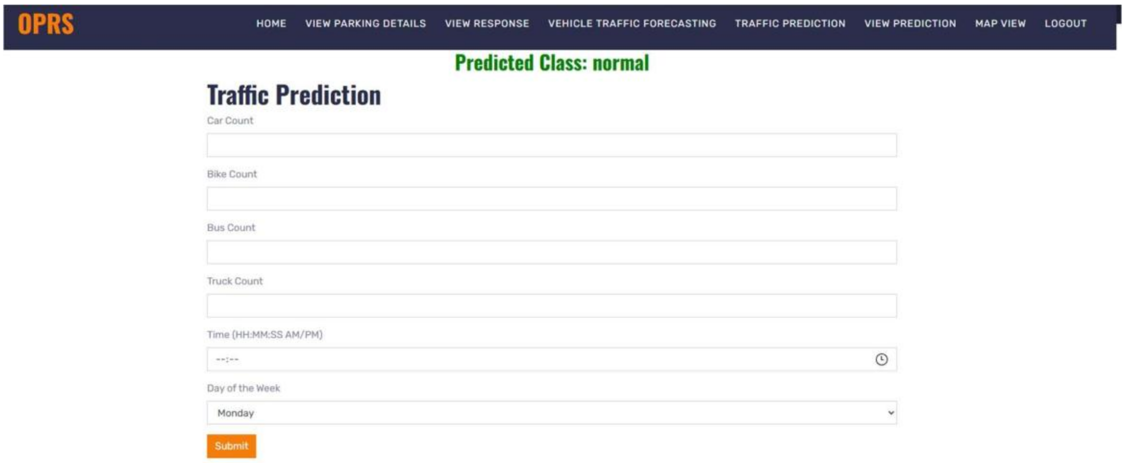


Hours	Forcasted Vehicle Count
1	145.0
2	129.0
3	120.0
4	109.0
5	101.0
6	101.0
7	105.0
8	109.0
9	109.0
10	122.0
11	140.0

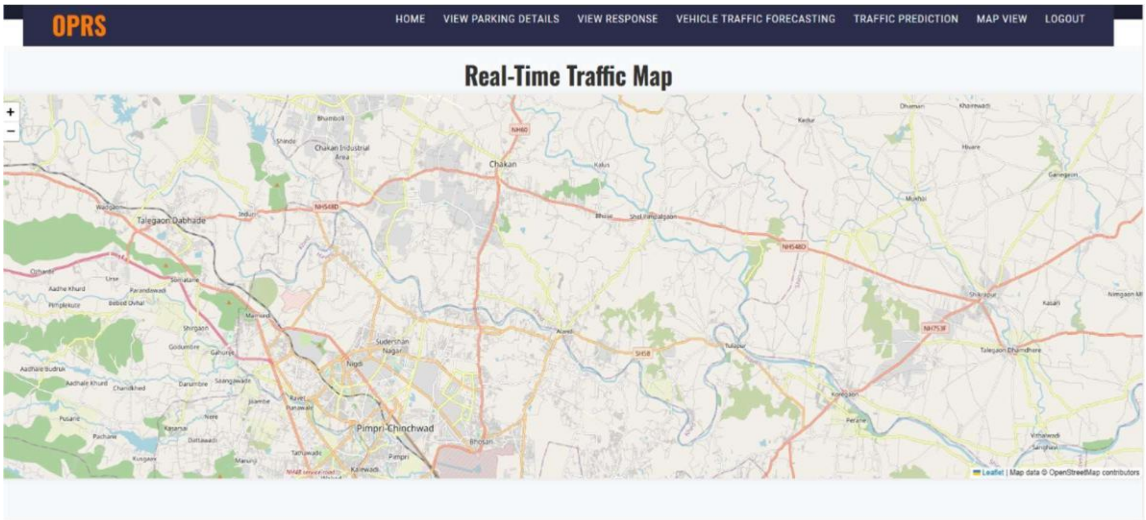
Traffic flow prediction: Visual representation of estimated vehicle flow on different routes.



Traffic prediction Response: Visual representation of estimated vehicle flow on different routes output.



Map view: Interactive map showing parking locations and real-time traffic data.



Traffic prediction through video data: Video-based interface analyzing real- time traffic for congestion prediction.

Output: Final screen summarizing predictions and analysis from all modules.